IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE PART IV: MEng and ACGI

HVDC TECHNOLOGY AND CONTROL

Tuesday, 14 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Questions 1 & 2 carry 25 marks each, Question 3 carries 30 marks

Any special instructions for invigilators and information for candidates are on page 1.

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Answer ALL Questions

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Question 1

a) Consider a bipole LCC HVDC link and an equivalent three-phase AC transmission line having the same conductor and insulator ratings. Derive the reason why the active power transfer capacity of both are approximately the same.

[5]

b) For an LCC-HVDC, the presence of low-order harmonics on both the AC side and the DC side of the converter is a concern. How is this problem mitigated with a 12-pulse converter arrangement? There is no need to draw the voltage or current waveforms or write the expressions for the voltage and current that involve the harmonics.

[5]

c) The direction of power flowing through an LCC HVDC link is reversed by changing the polarity of the direct voltage. Describe how this is done in practice by sketching the combined rectifier-inverter characteristics of the converters at both ends.

[5]

d) The rectifier end of an LCC-HVDC link controls the direct current while the inverter end operates with a fixed extinction advance angle. A three-phase short circuit occurs within the AC system at the rectifier end of an LCC HVDC link. How would the converters at both ends react and potentially try to avoid the possible shutdown of the link?

[5]

e) Explain the process of commutation failure at the inverter of an LCC-HVDC link considering the transfer of current from thyristor 1 to 3 in a standard 6-pulse converter configuration. No diagram is necessary. Under what situation is commutation failure likely to occur and how can it lead to successive commutation failures?

[5]

Question 2

a) An LCC-HVDC link is planned between two buses of a 400 kV AC system. The equivalent impedances of the AC system looking at those buses are 3.5Ω and 5.5Ω. The maximum reactive power generated by the capacitors and filter banks at each converter station under nominal bus voltage is 60% of the rated active power capacity of the LCC-HVDC link. Calculate the maximum power rating (in GW) of the LCC-HVDC link if the effective short circuit ratio must be at least 4.25.

[5]

b) The LCC-HVDC link in part (a) is built with a direct voltage rating of ±800 kV and uses a standard 12-pulse converter arrangement in each pole. The converter parameters are given in Table 2.1.

Nominal gate delay angle	17°
Minimum gate delay angle	2.5°
Minimum extinction advance angle	16°
Resistance of each pole	6 Ω
Commutation resistance for each 6-pulse converter	8 Ω
Current margin	16%

Table 2.1: Parameters of the converter at each end and the LCC-HVDC link

Assume that the rectifier end controls the rated current while the inverter end maintains the rated direct voltage under normal operation. Using the calculations on a per pole basis, answer the following questions:

i) Under normal condition with rated voltage on the AC system side of the converter transformers at both ends, determine the effective turns ratio of the converter transformer at each end.

[8]

ii) Determine the effective turns ratio of the converter transformers at each end when the voltage on the AC system side of the rectifier end converter transformer reduces by 10% starting from the normal condition.

[8]

iii) Determine the maximum gate delay angle of the rectifier immediately after the voltage on the AC system side of the inverter end converter transformer reduces by 10% starting from the normal condition.

[4]

Question 3

a) Consider a basic one-arm modular converter connected between an AC grid and a DC grid represented as equivalent voltage sources as shown in Figure 3.1. The parameters of the converter are given in Table 3.1. Answer the questions below.

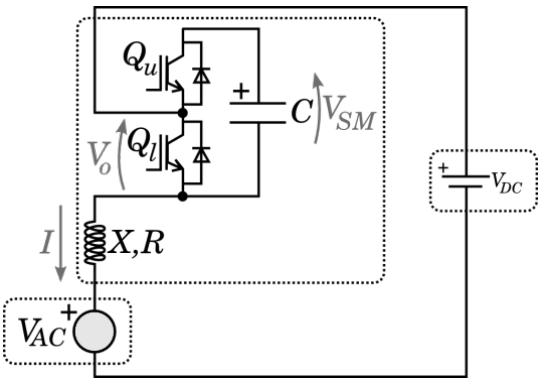


Figure 3.1 – Diagram of a one-arm modular AC/DC converter

Parameter	Value	Units	Brief description
V_{DC}	500	V	Rated DC grid voltage.
V_{AC}	230	V rms	Rated AC grid voltage
f_{rated}	50	Hz	Rated frequency of the AC grid
R	0.3	Ω	Equivalent resistance of the bridge plus the filter
X	2.5	Ω	Reactance of the filter
V_{SM}	1,000	V	Submodule capacitor voltage
H_C	40	ms	Equivalent inertia time of the submodule capacitor
P_{rated}	3,000	W	Rated active power

Table 3.1 – Parameters of the one-arm modular AC/DC converter

i) Write the differential equation that describes how the current of the circuit, I, is affected by the grid voltages, V_{DC} and V_{AC}, and the output voltage of the submodule, V_o. Apply the Laplace transform to the equation and find the transfer function between V_o and I.

[4]

ii) Answer the following questions related to the steady-state analysis of the converter:

a. Calculate the capacitance of the submodule, C.

[2]

b. What voltage would the submodule apply in the steady state, V_o, to operate at zero power?

[2]

c. Calculate the AC and the DC components of the voltage the submodule would apply, V_o, and the current in the circuit to deliver rated active power to the AC grid (and zero reactive power) while maintaining zero net power balance across the submodule.

[5]

iii) Answer the following questions related to the design of the controllers of the converter:

a. Would a PI controller be an appropriate choice to control the current of the converter? Justify your answer.

[2]

b. Would the voltage of the submodule capacitor, V_{SM}, have a ripple under steady-state operation? If this is the case, how should this be taken into account when designing the controller of the V_{SM} (or the controller of the energy of the submodule)? Justify your answer.

[3]

b) Answer the following question related to the AC-DC converter shown in Figure 3.2. The converter parameters are summarised in Table 3.2.

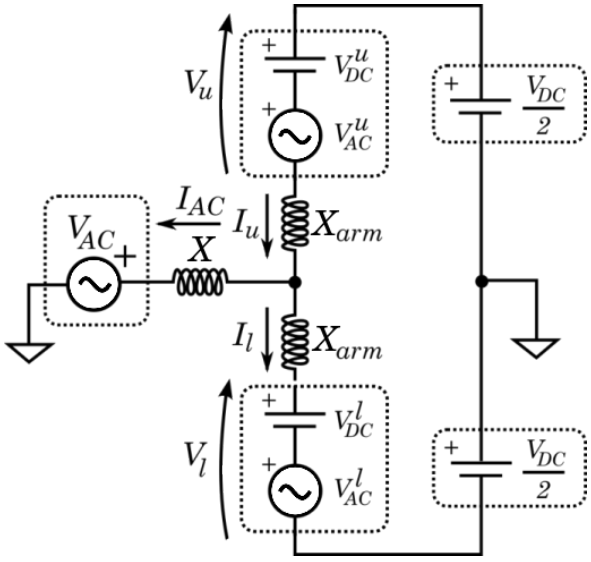


Figure 3.2 – Equivalent circuit of a two-arm single-phase AC/DC modular multilevel converter

Parameter	Value	Description
V_{DC}	1400 kV	DC network voltage.
V_{AC}	400 V	AC network voltage (RMS).
f_{AC}	50 Hz	Frequency of the AC network
S_{rated}	200 kVA	Rated power.
X_{arm}	0.25 Ω	Arm reactance.
X_L	0.05 Ω	Phase reactance
R_{arm}, R_L	0 Ω	Equivalent series resistances

Table 3.2 – Parameters of the two-arm single-phase AC/DC modular multilevel converter.

i) Consider the converter operating at zero power (zero current everywhere) and answer the following questions:

a. Calculate the AC and the DC components of the output voltage of the stacks, V_u and V_l,

[2]

b. What is the minimum number of submodules required for each arm of the converter to be able to operate at zero power if the voltage of an individual submodule is 700 V?

[3]

ii) What current would the converter use if it were to draw 20 kW from the upper pole of the DC-link and 10 kW from the lower pole of the DC-link at a given point in time? Would this operating point be used in practice? Briefly explain why.

[3]

iii) Consider a transient operating point where the power exchanged with the AC grid was zero but the energy controller of the converter required 10 kW to be drawn from the upper stack and delivered to the lower stack. Calculate the AC and DC voltages and currents of the converter for this operating condition. State any approximations made.

[4]